



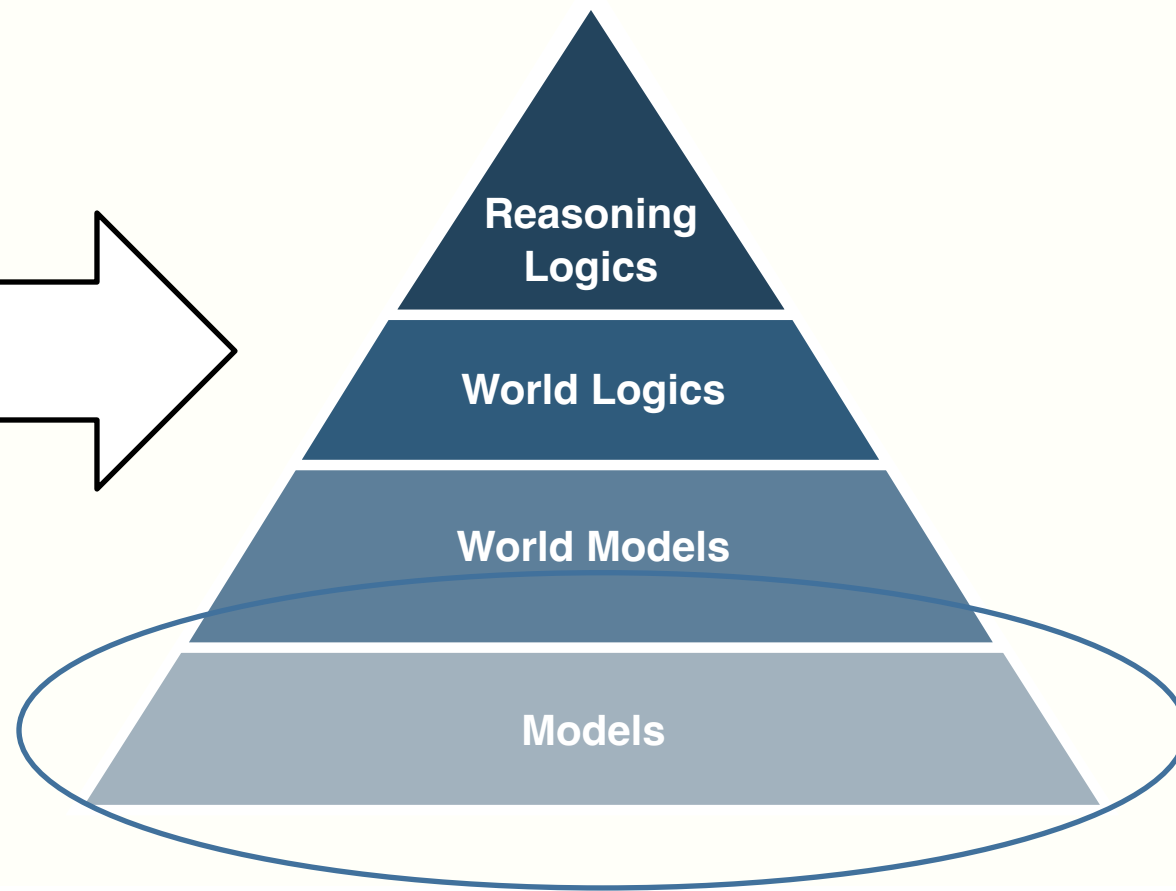
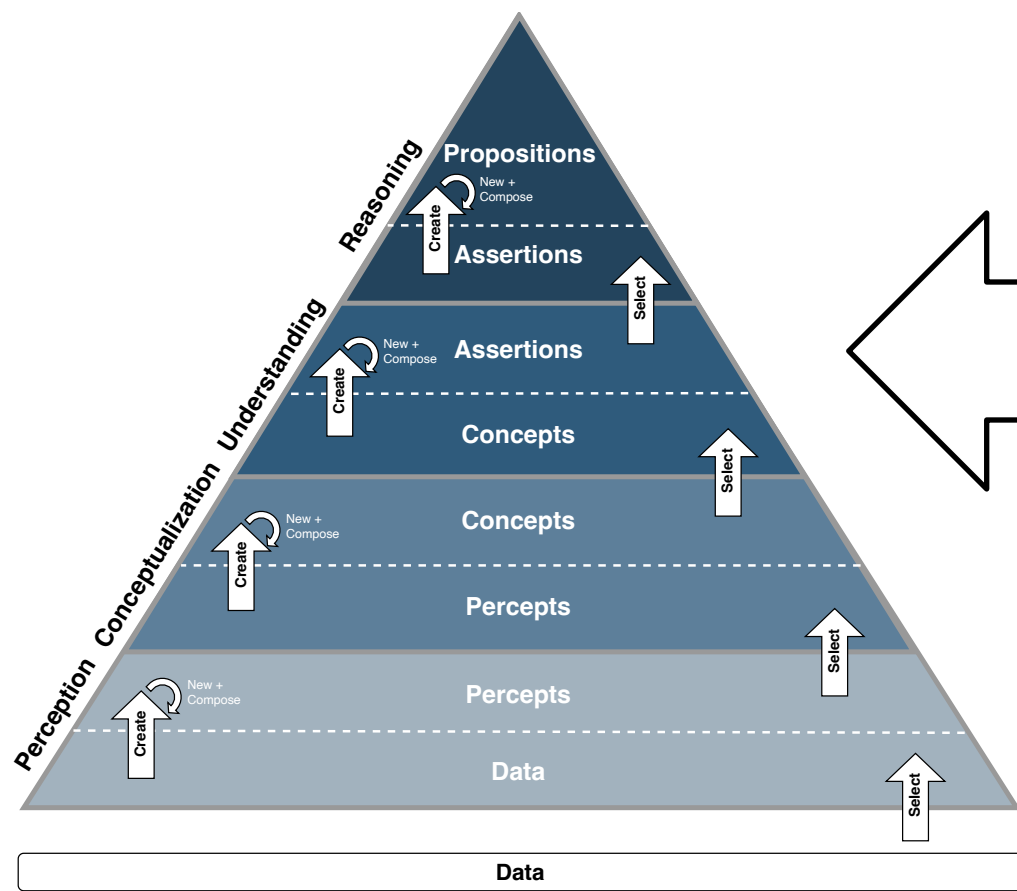
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Dipartimento di Ingegneria e Scienza dell'Informazione



Models (of the world) (HP2T)

The knowledge Pyramid and the Logics pyramid (recall)

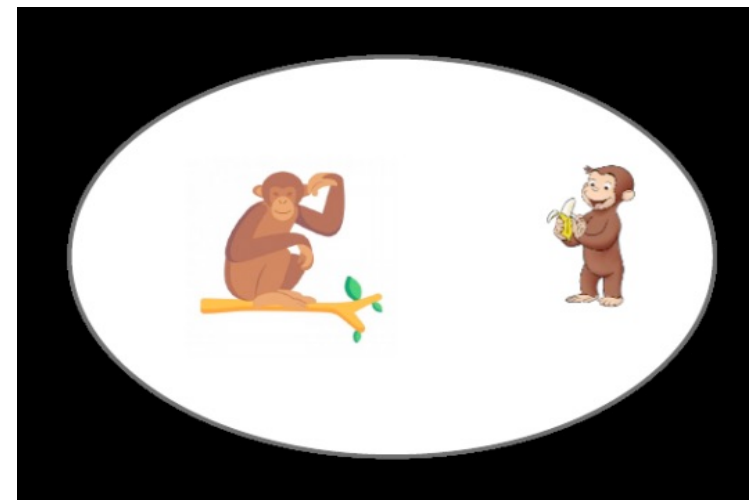


Index

- **Intuition**
- Things, entities
- Percepts
- Domains
- Facts
- Models
- Key notions

Analogic representations

- There is a tree
- There is a banana
- The monkey is eating a banana
- The monkey is sitting on a tree
- The monkey is scratching its head



Observation 1. Which monkey? There are two monkeys! The language used in the linguistic representation is ambiguous! At the same time there is general way to “link” the language descriptions of the linguistic representation to the element of the analogic representation.

Observation 2. Preliminary to any attempt of disambiguation of representations we need to make precise what analogic representations depict.



Index

- Intuition
- **Things, entities**
- Percepts
- Domains
- Facts
- Models
- Key notions

Analogic representations – Things

Observation (What can be seen in a picture: things). The core element of perception is **things**, where:

- Any thing can be clearly **distinguished** from any other thing, e.g., people, objects, dogs, hair;
- There cannot be two things in the **same** spatio-temporal coordinates;
- A thing moves **smoothly** in space-time (facilitates recognition in time);
- We see them multiple times, from **occurrences** (of things) to **things**, modulo ability to recognize two occurrences to be the same thing;
- Things are all **different** from one another (there are no two things which are identical);
- Things are all in **different** mutual relations.



We always perceive **multiple things**, that we can distinguish from one another, grounded in their space-time coordinates

Addendum* - Thing and Substance (Philosophy)

“There exist two rather different ways of characterising the philosophical concept of substance. The first is the more generic. The philosophical term “substance” comes from an early Latin translation of the Greek *ousia*. *Ousia* is a noun derived from the verb “*einai*” (to be) and is naturally translated “being”. According to the generic sense, substances are those things that best merit the title “*beings*”. This is usually interpreted to mean those things that are the foundational or fundamental entities of a given philosophical system. ... The second use of the concept is more specific. According to this usage, substances are a particular kind of entity, and some philosophical theories acknowledge them and others do not. This conception of substance derives from the intuitive notion of individual thing or object, which contrasts mainly with properties and events. According to this usage, it is a live issue whether the fundamental entities are substances or something else, such as events, or properties located at space-times. The issue is how we are to understand the notion of an object, and whether, in the light of the correct understanding, it remains a basic notion, or one that must be characterised in more fundamental terms. Whether, for example, an object can be thought of as nothing more than a bundle of properties, or a series of events.” [*]

“... *substances* are those things about which you can learn from one encounter something of what to expect on other encounters, where this is no accident but the result of a real connection.” [**]

“... *substance concepts* are taken to be *recognition abilities*, namely abilities which allow a *perceptual-cognitive system* to realize that the substance involved in the current encounter is the same substance as from previous encounters.” [***]

[*] *Stanford Encyclopedia of Philosophy*. <https://plato.stanford.edu/entries/substance>

[**] Millikan, R. G. (2000). *On clear and confused ideas: An essay about substance concepts*. Cambridge University Press.

[***] Giunchiglia, Fausto and Fumgalli, Mattia. Concepts as (recognition) abilities. In: *Formal ontology in information systems*. IOS Press, 2016. p. 153-166.



Analogic representations – from Things to Entities

Observation (From things to entities). We call **entity** a single **thing** that we recognize as such, e.g., Sofia, a dog;

- The **specific entity** being recognized from the thing we perceive is largely **arbitrary**. For instance, what do you see in the picture on the right: an organism, an organism which is alive, a person, a woman, a black person, a black woman, Sofia, a friend, an enemy, a victim to kill, ...?
- The first motivation underlying the selection of a specific entity is the **partiality** of the input. For instance: did you see the thing from the back, from far away, for a second, ...?;
- The second motivation is your **background** (previous knowledge, memory): you are an alien, a social scientist analyzing a specific social phenomenon, a suprematist, a friend, a killer ...?
- The third main motivation is the current **purpose**. For instance: you are looking for a form of life, for a friend, for Sofia?
- Partiality, background and purpose are elements of the current **context (external, internal, motivational)**.



Our perception of an entity out of a thing does not change, it is **stable**, and as **specific as possible**, given the current context. **But changes across contexts.** This is the main source of **representation diversity**.

Analogic representations – Entities

Observation (Entities). We have the following:

- **Entities**, that is, what we perceive from **things**, e.g., Sofia, a dog;
- **Entity types**, also called **etypes**, that is, **multiple entities which somehow look the same**, e.g., the etype woman, dog, tree;
- **Entity properties**, that is, **entities with certain characteristics**, e.g., a woman with blond hair, a dogs barking;
- **Property types**, that is, **multiple entities with certain characteristics**, similar among them, e.g., two women with blond hair, two dogs barking;
- **Entity Relations**, that, is **multiple entities in some relation among them**, e.g., a woman talking to a man with a phone, a dog between a woman and a man, a woman friend of a man;
- **Relation types**, that is, **groups of multiple entities in some relation among them**, similar among them, e.g., two groups of a man and woman talking to one another with a phone, two couples woman and man, both with a dog in between.



Relations and properties are **identified after the identification** of the entity: Do you see a woman which is black (together with a white one) or a black woman?

Analogical representations – Sameness vs. difference

Observation (Entity sameness). Entities are identified by a set of properties which always hold about them. For instance, a black person is no longer a black person if it changes the color of his/he skin.

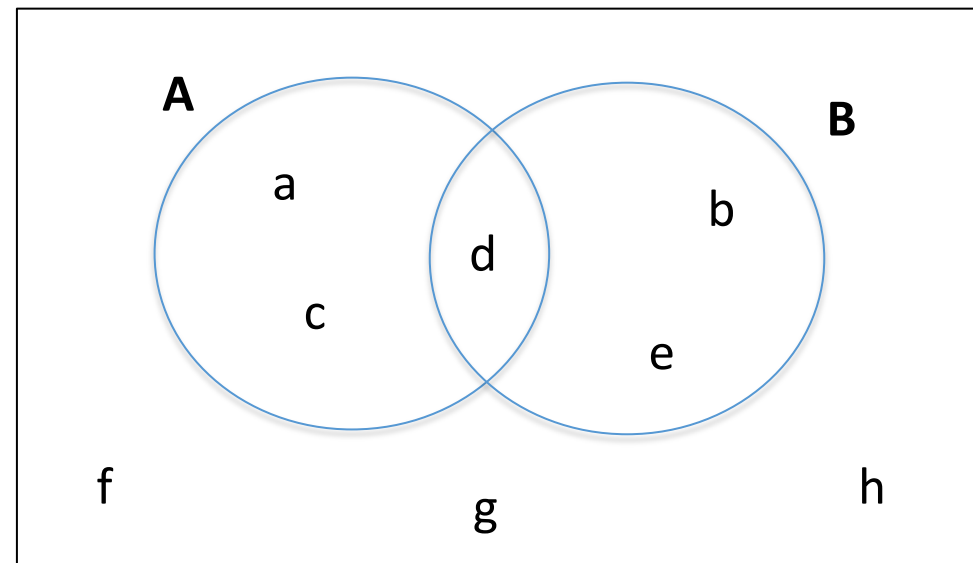
Observation (Entity difference). What makes an entity identical to itself (more or less, from your selected point of view) makes it different from every other entity:

- We must **distinguish** any **entity occurrence** from any other entity occurrence;
- Given **two distinct occurrences of two entities**, we must decide whether they are occurrences of the **same** entity or of two **different** entities (the two entities must be different if perceived at the same time);
- Given **two distinct entities**, we must decide whether they are members of the **same etype** or of two **different etypes**;
- Given **two distinct entity tuples**, we must decide whether they are the **same tuple** or of two **different tuples**;
- Given **two distinct entity tuples**, we must decide whether they are members of the **same property type or relation type** or of two **different property types or relation types**.



equality / inequality
vs.
similarity / dissimilarity
vs.
sameness / difference
of entities

An analogical representation -What do you see?



Intuition (Analogic representations in set theory). We model analogic representations using **sets**, where things are modeled as **single elements** and groups of things with properties or relations in common are modeled as **sets**.

Why set theory?

- It is a **linguistic representation describing** (partially, and **NOT** depicting) selected features of an analogic representation,
- It has a linguistic representation (set-theory) and an analogic representation (Venn Diagrams)
- There is a *one-to-one mapping* between any set-theoretic representation of the world and Venn Diagrams
- It is *universal* – it can be used to represent ANY real world situation. Via: set **membership**, set **inclusion**, set **equality**, set of **tuples**.
- It is *self-evident* – it formalizes the **similarity** and **dissimilarity** (as **set equality/inequality**), i.e., what is different from what and what is similar to what,
 - of **entities** and **etypes**
 - of **entity properties** and **property types**
 - of **entity relations** and **relation types**
- It has a formal language (type 1 language)

Set theory – Terminology

Terms naming components of linguistic representations

- *Entity*
- *Etype*
- *Entity property*
- *Property type*
- *Entity relation*
- *Relation type*
- ???
- ???
- ???
- ???

Corresponds to

Set-theoretic terms naming components of analogic representations

- *Unit, element*
- *Class / set*
- *Tuple*
- *Relation, that is set of tuples*
- *Tuple*
- *Relation, that is set of tuples*
- ???
- ???
- ???
- ???

Each terms on the left (e.g., entity) is the name of the element which plays the same role as the element whose name is in the same line, on the right (e.g., unit).

Notation (terms used). In informal writing, whenever no confusion arises we will use terms on the right in place of terms on the left, to facilitate the interpretation.

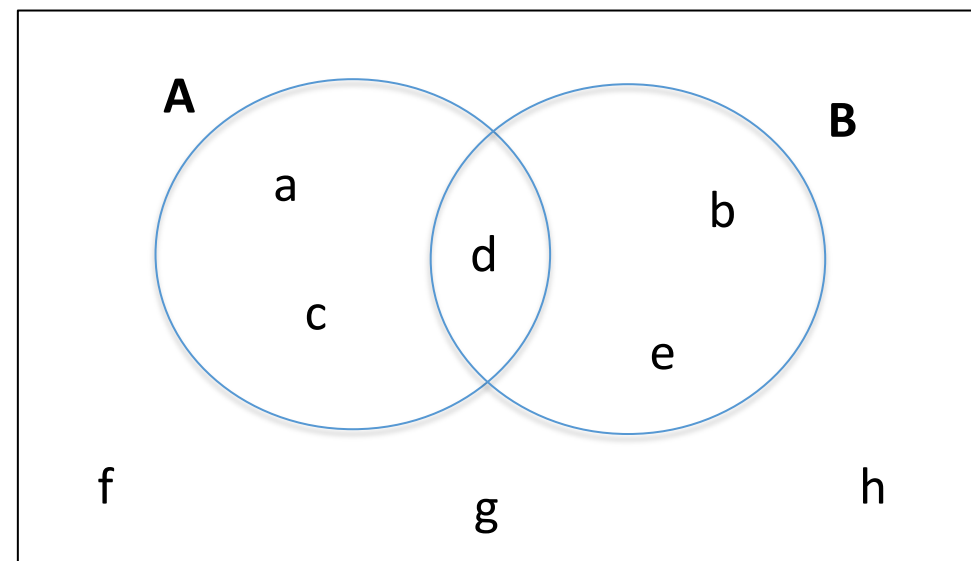
Index

- Intuition
- Things, entities
- **Percepts**
- Domains
- Facts
- Models
- Key notions

An analogical representation -What do you see?



Observation 1. Analogic representations are formalized using set theory



Observation 2 (Percepts, domain). Percepts are all the things or combinations of things which can be distinctively perceived, inside the selected **domain** (the box).

Example (Domain). The set of percepts of the picture on the left.

Percepts – Intuition

Intuition (percept) A **percept** is an entity or a combination of entities (a perception) which is perceived as distinct from others.

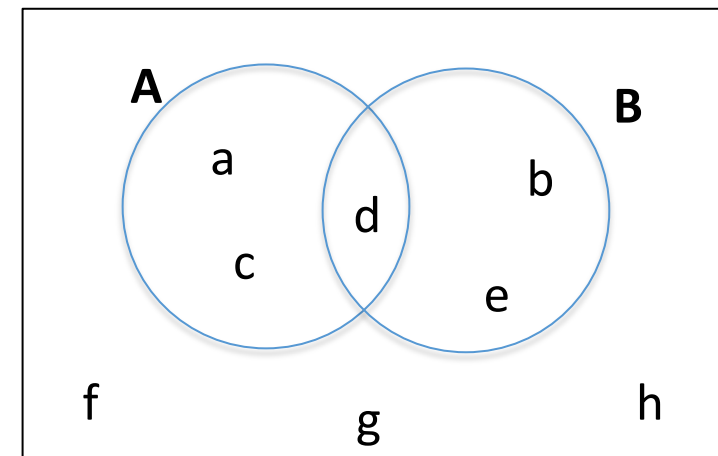
Examples: The percepts described by the sentences below

1. An *entity* you know (e.g., *Rocky*) or don't know (e.g., *woman#1*)
2. The *etype* *People* populated by entities who are persons (an etype)
3. The *property* *blond* holding of a *blond* entity (e.g., *person#1*)
4. The *property type* *blond* holding of the people who are blond (a property of a subset of the etype *people*)
5. The *relation* *near* holding of Sofia when *near* Paolo (a relation between entities)
6. The *relation type* “*talking to*” holding of *women talking to men* (a relation between two subsets of the two etypes *woman* and *man*)

Percepts - definition

Definition (percept) A **percept** is any of the six cases below.

- **Entities**, $e_1, \dots, e_n$, perceived as distinct *units* (in space and time, e.g., Sofia);
- **Entity properties**, $P_1, \dots, P_n$, perceived as *relations* between entities (e.g., Sofia is blond)
- **Entity relations**, $R_1, \dots, R_n$, perceived as *relations* among entities (e.g., Rocky is between Sofia and Mark)
- **Etypes**, $E_1, \dots, E_n$, as *classes* of entities, (e.g., person)
- **Property types**, $P_1, \dots, P_n$, as *classes* of tuples of entities (e.g., Swedish people are blond)
- **Relation types**, $R_1, \dots, R_n$, as *classes* of tuples of entities (e.g., breaks happen between lectures)



Example –Which percepts do you see?



Observation. The strings used in the text of the example are **pointers** to a region in the analogic representation. They are **NOT** words in some language

Observation (Which percepts can be seen in a picture). We have the following percepts (those which are underlined)

Entities

Rocky, tree#1, person#12, Sofia, animal#1, hair#1, hair#2, entity#1,...

Entity types (etypes)

Person = {Sofia, Paolo, woman#1, ...}

Entity properties

< hair#2, black>, < hair#1, blond>, ...

Property types

haircolor = {<hair#2, black>, <hair#1, blond>, ...}

from which we can write:

< hair#2, haircolor black>, < hair#1, haircolor, blond>

Entity relations

<Rocky, Sofia, Paolo>, <Sofia, tree#1, Paolo>, ...

Relations types

between = {<Rocky, Sofia, Paolo>, < Sofia, tree#1, Paolo >, ...}

from which we can write:

<between, Rocky, Sofia, Paolo>,

Index

- Intuition
- Things, entities
- Percepts
- **Domains**
- Facts
- Models
- Key notions

Domain of interpretation - intuition

Observation (Percepts within a boundary). A **percept** is anything which is perceived as distinct from others. But perceived when and where? We need to define a boundary within which we select what we perceive.

Intuition (Domain of interpretation). We call **Domain of interpretation** the set of percepts that is within the scope of what we consider. That is, if **D** is a Universe of interpretation and **p** is a percept, then

$$D = \{p\}$$

where **{p}** is the set of selected percepts.

Observation (Different types of percepts). We can distinguish percepts based on our analysis of what we perceive.

Domain of interpretation - definition

Definition (Domain of interpretation) A Domain (of interpretation) D is defined as

$$D = \langle U, \{C\}, \{R\} \rangle$$

where:

- $U = \{u\}$ is called the **Universe (of interpretation)** of D .
- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, for some n
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of elements, for some m , with $C_i \subseteq U$
- $\{R\}$ is a set of n -ary **relations** $R_1^n, \dots, R_p^n$ among elements, for some p , with $R_i^n \subseteq U \times \dots \times U$
- $u_1, \dots, u_n, C_1, \dots, C_m, R_1^n, \dots, R_p^n$ are **percepts**.

Domain of interpretation - intuition

Observation (Diversity of percepts). From a set theoretic point of view we have three different types of percepts:

- (i) units: u
- (ii) classes: C
- (iii) relations: R^n

where:

- Units depict entities
- Classes depict sets on entities
- Relations depict properties of sets of entities
- Relations depict relations of sets of entities

Index

- Intuition
- Things, entities
- Percepts
- Domains
- **Facts**
- Models
- Key notions

Facts - Intuition

Intuition (Fact) A **fact f** is a **perception** happening at certain spacetime coordinates.

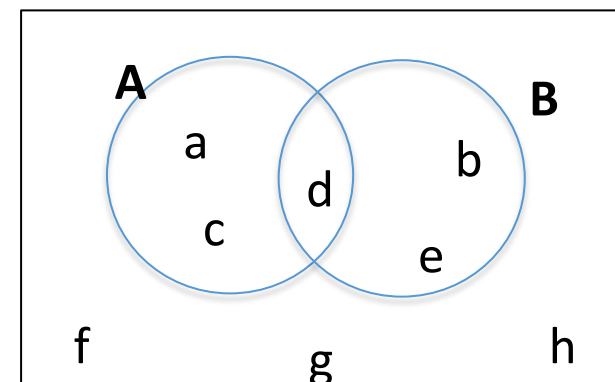
Example (fact). Facts involving percepts and facts denoted by the sentences below

- *Spacetime invariant facts*, e.g., dogs are animals, a woman is a person
- *Time invariant facts*, e.g., there is a church in Trento near the square
- *Space invariant facts*, e.g., moving across continents requires flying
- *Spacetime variant facts*, e.g., Sofia has blond hair, Sofia is a friend of Paolo, Sofia is walking, Paolo is talking to Sofia

Facts – intuition (continued)

Observation 1 (Fact in set theory). In set theory a **fact** is modeled as a **relation** between percepts. All and only the facts are as follows

- A **MemberOf** relation
 - between a unit and a class
 - between a tuple of units and a relation
- A **SubsetOf / SupersetOf** relation
 - between two classes
 - between two relations
 - between a n-ary relation and a tuple of n classes



Example –Which facts do you see?



Observation (Which facts can be seen in a picture). We have the following facts:

Element memberOf class

woman#1 $\in$ Person

Tuple of elements memberOf relation

< Paolo, Rocky > $\in$ HasDog

Class subsetOf class

Woman $\subseteq$ Person

Relation SubsetOf relation

near $\subseteq$ friendOf

Tuple of classes subsetOf relation

Person x Dog $\subseteq$ HasDog

Relation subsetOf tuple of classes

HasDog $\subseteq$ Person x Animal

Types of facts

Definition (Fact). A **fact f** has one of the following five forms

- *Unit memberOf Class:* $u_i \in C_j$,
- *Tuple of Units memberOf relation:* $\langle u_1, \dots, u_n \rangle \in R^n$,
- *Class subsetOf Class:* $C_i \subseteq C_j$,
- *Relation subsetOf relation:* $R_i^n \subseteq R_j^n$
- *Relation subsetOf tuple of classes and viceversa:*
 - $R^n \subseteq C_1 \times \dots \times C_n$
 - $C_1 \times \dots \times C_n \subseteq R^n$

with: $u_i \in U$, $C_i \subseteq U$, $R^n \subseteq U \times \dots \times U$.

Example –Which facts do you see?



Observation. Percepts define the space of possible facts. **Facts** are (representations of) **perceptions** which describe what is the case. Facts are relations/ properties relating entities or sets of entities to one another. Elements alone cannot be facts!

Observation (Which facts involving percepts can be seen in a picture). We have the following (yellow)

Entities (Elements)

Rocky, tree#1, person#12, Sofia, animal#1, Thing#1,...

Entity properties (Element binary relations)

<haircolor, hair#2, black>, < haircolor,hair#5,brown>, ...

Entity relations (Element n-ary relations)

<between, Rocky, Sofia, Paolo>,
<between, Sofia, tree#1, Paolo>, ...

Etypes (Classes)

Person = {Sofia, Alfredo, Martin, ...}

Etype properties (Class binary relations)

color = {<hair#2, black>,<hair#1,blond>}

Etype relations (Class n-ary relations)

between = {<Rocky, Sofia, Paolo>,< Sofia, tree#1,
Paolo >, ...}

Facts and domains

Observation 1 (percept, domain, fact). As defined before we have that a domain **D** is defined as a set of percepts, that is $D = \{p\}$. From the definition of fact we can also see **D** as the set of facts, that is

$$D = \{f\}$$

where a fact $f \in \{f\}$ is built from percepts by applying any of the equations used in the construction of fact.

Observation 2 (percept, domain, possible fact). The construction of a **domain** depends on two modeling choices:

- The selection of the set of **percepts** $\{p\}$
- The selection of the set of **possible facts** $\{f\}$. Domains are usually assumed to contain all **possible facts** which can be composed from a selected set of percepts $\{p\}$.

where by “possible fact” we mean a fact (relation among percepts) which may eventually be what is the case

Index

- Intuition
- Things, entities
- Percepts
- Domains
- Facts
- **Models**
- Key notions

Models - Intuition

Intuition (Model) A **model** is what you perceive, that is the set of facts you perceive, not necessarily all of them (within a certain spectime boundary).

Example: A possible model depicting the picture above is the following set of facts:

$$\{Sofia \in person, Paolo \in man, Rocky \in dog, \\ \langle Sofia, Paolo \rangle \subseteq near, \langle Sofia, hair\#32 \rangle \in blond, \\ \langle Sofia, Paolo \rangle \in friendOf, man \subseteq person \\ Rocky \in animal, \langle Rocky, Sofia \rangle \in dogOf, \dots\}$$

Observation 1 (Which facts in a model?). Because of its partiality, a model can be extended by adding new facts. For instance, the example model above could be extended by asserting the fact "*Sofia $\in$ woman*".

Observation 2 (Which facts in a model?). **NOT all facts can be added to a model!** For instance, in the model above one can NOT add the fact "*Paolo $\in$ woman*". A model cannot contain two facts where the presence of one excludes the presence of the other for the model to depict a possible situation in the real world.

Models - definition

Intuition (Mutually inconsistent facts). Two facts f_1, f_2 are **(mutually) inconsistent** if $f_1 \in M$ implies the impossibility of $f_2 \in M$, and vice versa.

Definition (Model). A set of facts $\{f\}$ is a **model**

$$M = \{f\}$$

if $\{f\}$ does not contain two inconsistent facts.

Terminology (Inconsistency, consistency). If $f_1 \in M$, and if f_1, f_2 are two inconsistent facts then we say that f_2 is **inconsistent with** M and that $\{f_2\} \cup M$ is **inconsistent**, and that $\{f_2\} \cup M$ is **not a model**. Two models are (mutually) **inconsistent** if they contain two (mutually) inconsistent facts. **Consistency** means absence of inconsistency.

Facts, models and domains

Proposition (Domain and facts). A Domain D is a set of facts $\{f\}$.

$$D = \{f\}$$

Proposition (Model). Given a domain D , a model M in D is a subset of D .

$$M = \{f\} \subseteq D$$

Observation (Domain, model). A domain is the set of all facts that we are willing to consider. A model is just the subset of facts that we define when depicting what is the case in the current situation.

Facts, models and domains

Observation (Domain) A Domain defines all and only what can be potentially perceived. It can be thought of as the set of all the possible things and also as the set of all the possible models.

Observation (Inconsistent facts in a domain). A domain, differently from a model, may contain any number of **inconsistent facts**. Given a domain, there are many potential models, some of which are potentially mutually inconsistent.

Observation. Domains must allow for the possible instantiation of distinct mutually inconsistent models, as it is normally the case in the world.

Model theory (terminology)

Observation (Model theory) . Model theory, that is, the theory of models, is the discipline that studies the structure of models as a function of specific types of percepts and facts.

In model theory, models are formalized as algebraic structures

Set theory – Terminology

Terms naming components of linguistic representations

- *Entity*
- *Etype*
- *Entity property*
- *Property type*
- *Entity relation*
- *Relation type*
- *???*
- *???*
- *???*
- *???*

Corresponds to

Set-theoretic terms naming components of analogic representations

- *Unit, element*
- *Class / set*
- *Tuple*
- *Relation, that is set of tuples*
- *Tuple*
- *Relation, that is set of tuples*
- *Percept*
- *Domain of interpretation*
- *Fact*
- *Model / analogic representation*

Each terms on the left (e.g., entity) is the name of the element which plays the same role as the element whose name is in the same line, on the right (e.g., element).

Notation (terms used). In informal writing, whenever no confusion arises we will use terms on the right in place of terms on the left, to facilitate the interpretation.

Index

- Intuition
- Things, entities
- Percepts
- Domains
- Facts
- Models
- **Key notions**

Key notions

- Formalization of analogical representations as model theory
- Model theory as set theory
- Thing
- Representation diversity (leading to subjectivity, ...)
- Entity
- Percept
- Domain
- Fact
- Model
- Inconsistent facts, models
- Correlations between percepts, facts, models and domains
- Model theory



Models (of the world) (HP2T)